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5.1:

11.3.28

Assignment - 5

A1. a)

$(v_o - v_x) s C_c = v_x \times s C_i$
 $\therefore v_o = v_x \left(\frac{C_i + C_c}{C_c} \right)$

$G_{m2} V_T + (v_o - v_x) s C_c + v_o [s C_o + s C_L + s C_x] = 0$
 $\therefore G_{m2} V_T = -v_x \left[s C_i + \left(\frac{C_i + C_c}{C_c} \right) (s C_o + s C_L + s C_x) \right]$

$\therefore \frac{v_x}{V_T} = L_2(s) = \frac{-G_{m2} C_c}{s [C_i C_c + (C_i + C_c)(C_o + C_L)] + C_x (C_i + C_c)}$

Pole of the closed loop transfer function,
 $f_z = \omega_{u, \text{loop}} \text{ of } L_2(s)$

$L_2(s) = \frac{A_0}{1 + \frac{s}{\omega}}$
 $|L_2(s)| = 1 \Rightarrow \frac{A_0^2}{1 + \left(\frac{\omega_{u, \text{loop}_2}}{\omega} \right)^2} = 1$
 $\therefore (\omega_{u, \text{loop}_2})^2 = \omega^2 (A_0^2 - 1) \approx \omega^2 A_0^2$

$\therefore \omega_{u, \text{loop}_2} = f_z = \omega A_0 = \frac{G_L (C_i + C_c)}{C_c (C_i + (C_i + C_c)(C_o + C_L))} \times \frac{G_{m2} C_c}{G_L (C_i + C_c)}$

$\therefore f_z = \frac{G_{m2} C_c}{C_c (C_i + (C_i + C_c)(C_o + C_L))} = 4 \times \omega_{u, \text{loop}}$

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$$\therefore \frac{g_{m2} c_c}{c_c \times g_{m2} + \left(\frac{g_{m2}}{\omega_T} + c_c \right) \left(\frac{g_{m2}}{\omega_T} + c_L \right)} = 8\pi f_0$$

($\because \omega_{u, \text{loop}} \approx 2\pi f_0$)

$$\therefore g_{m2}^2 + g_{m2} \left[c_L + 2c_c - c_c \omega_T \right] \omega_T + c_c c_L \omega_T^2 = 0$$

$8\pi f_0$

$$\therefore g_{m2}^2 - 27.577 \times 10^{-3} g_{m2} + 96 \times 10^{-4} = 0$$

$$\therefore g_{m2} = 0.040875, 0.23495$$

b)

$\therefore L(s) = \frac{-A(s)}{k+1}$

Assume $A(s) = \frac{(1-s/z_1) A_0}{(1+s/p_1)(1+s/p_2)}$

$$\therefore \omega_{u, \text{loop}} :- |L(s)| = 1 \Rightarrow 1 = \frac{(A_0)^2}{(k+1)} \frac{[1 + \omega_{u, \text{loop}}^2/z_1^2]}{[1 + \omega_{u, \text{loop}}^2/p_1^2][1 + \omega_{u, \text{loop}}^2/p_2^2]}$$

Assuming $\omega_{u, \text{loop}} \ll z_1, z_2$ and $\omega_{u, \text{loop}} \gg p_1, p_2$:-

$$1 = \frac{(A_0)^2}{(k+1)} \times \frac{1}{\omega_{u, \text{loop}}^2/p_1^2} \Rightarrow \omega_{u, \text{loop}} = \frac{A_0 p_1}{k+1}$$

Assuming $A_0 = \frac{g_{m1} g_{m2}}{c_{o1} c_{o2}}$ and $p_1 \approx \frac{c_{o1} c_{o2}}{g_{m2} c_c}$

$$\omega_{u, \text{loop}} \approx 2\pi f_0 = \frac{g_{m1}}{(k+1) c_c}$$

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$$\therefore G_{m1} = 2\pi f \phi (A+1) C_c = 1130.97 \times 10^{-6} S$$

$$G_{m1} = 1.13 mS$$

$$\rightarrow \frac{G_{m1}}{C_1} = \frac{1.13 \times 10^{-3}}{1.13 \times 10^{-3} + \frac{G_{m2}}{\omega_T}} = \frac{\omega_T G_{m1}}{G_{m1} + G_{m2}} \approx \frac{\omega_T G_{m1}}{G_{m2}}$$

$$\frac{G_{m2}}{C_2} = \frac{G_{m2}}{G_{m2} + C_L \omega_T} = \frac{\omega_T G_{m2}}{G_{m2} + C_L \omega_T} = \frac{\omega_T G_{m2}}{G_{m2} + 0.48}$$

For $G_{m2} = 0.040875$, $\frac{G_{m1}}{C_1} = 0.027 \omega_T$ ($G_{m2} = 0.078 \omega_T$)

For $G_{m2} = 0.2355$, $\frac{G_{m1}}{C_1} = 0.005 \omega_T$ ($G_{m2} = 0.33 \omega_T$)

~~Thus, both values for G_{m2} are permissible.~~

~~$G_{m2} = 0.2355$ gives less steady state error, since it gives a higher A_o .~~

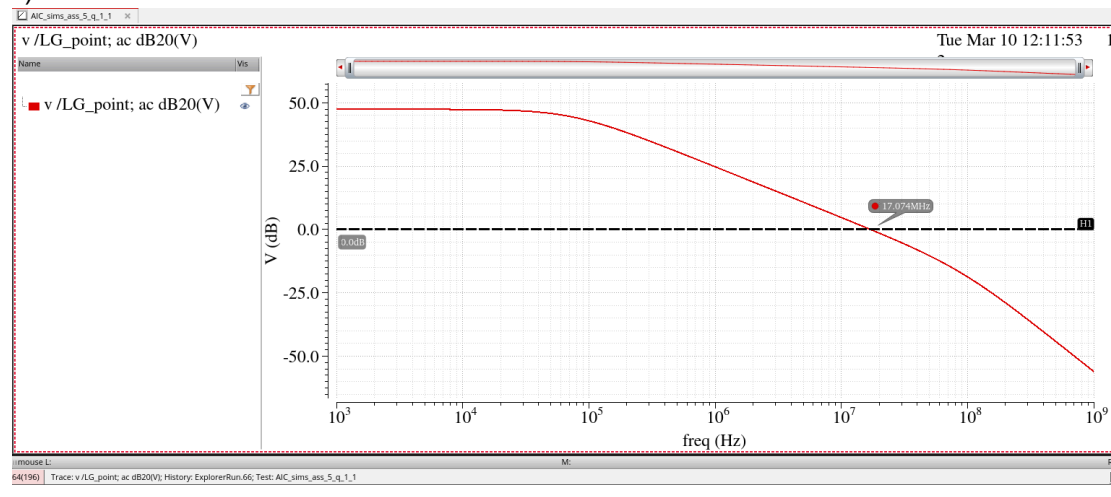
$G_{m2} = 0.04085$ has less power consumption ($\because$ power consumption $\propto G_m$, because the parasitics will be higher for higher G_m).

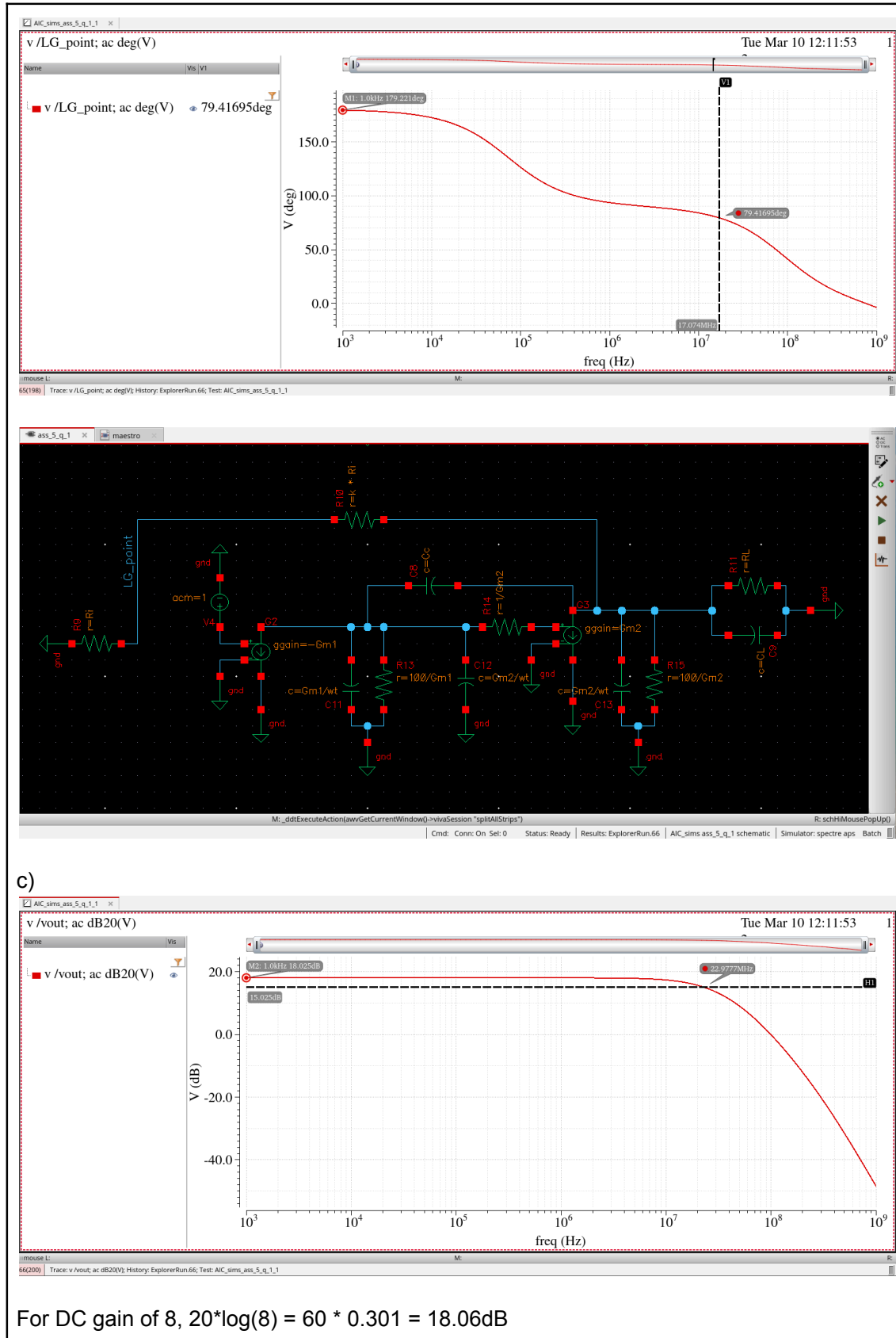
Thus, we will use $G_{m2} = 0.040875$ for further analysis.

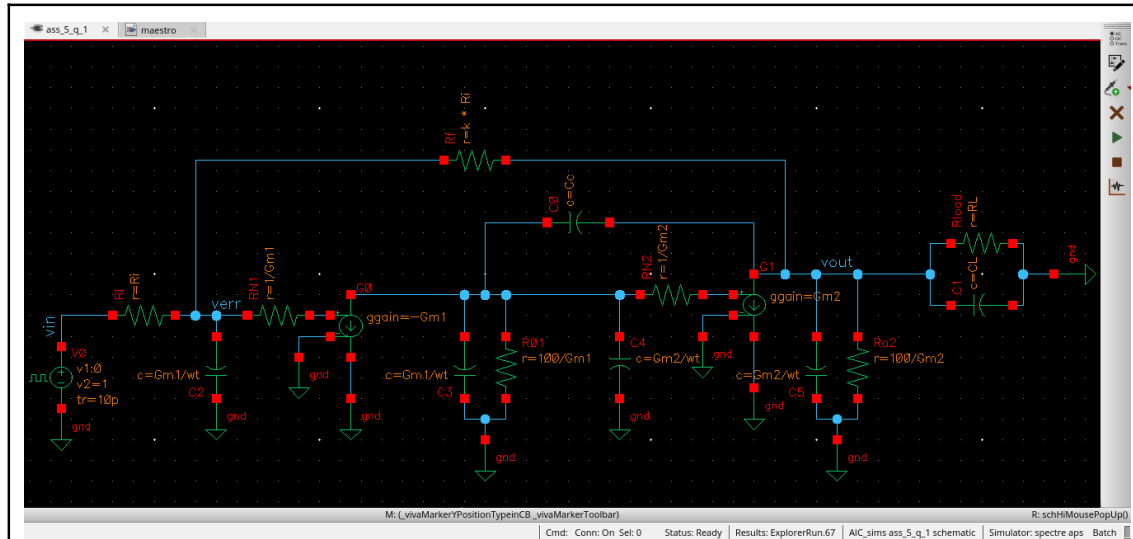
a)

Specification	Value	Unit
Closed loop dc gain k	8	-
Closed loop bandwidth f_B	20	MHz
Load capacitor C_L	24	pF
Load resistor R_L	663.14	Ω
Input resistance R_i	10	k Ω
G_{m1}	1.13	mS
G_{m2}	0.04087	S
$C_{i,1} = C_{o,1} = G_{m1}/\omega_T$	0.0565	pF
$C_{i,2} = C_{o,2} = G_{m2}/\omega_T$	2.0435	pF
$R_{o,1} = 100/G_{m1}$	88.495	k Ω
$R_{o,2} = 100/G_{m2}$	2.447	k Ω
$R_{N,1} = 1/G_{m1}$	884.95	Ω
$R_{N,1} = 1/G_{m2}$	24.47	Ω
C_c	1	pF

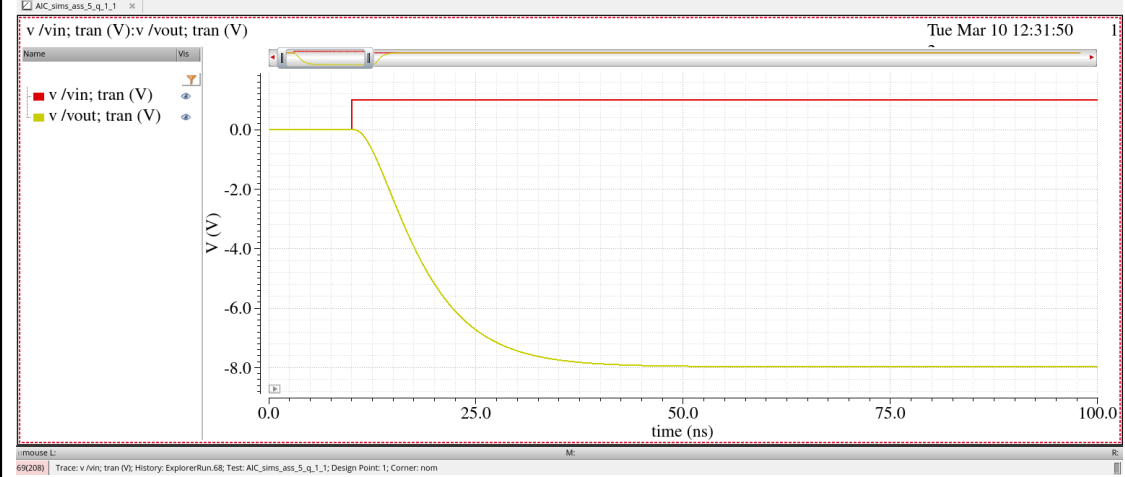
b)



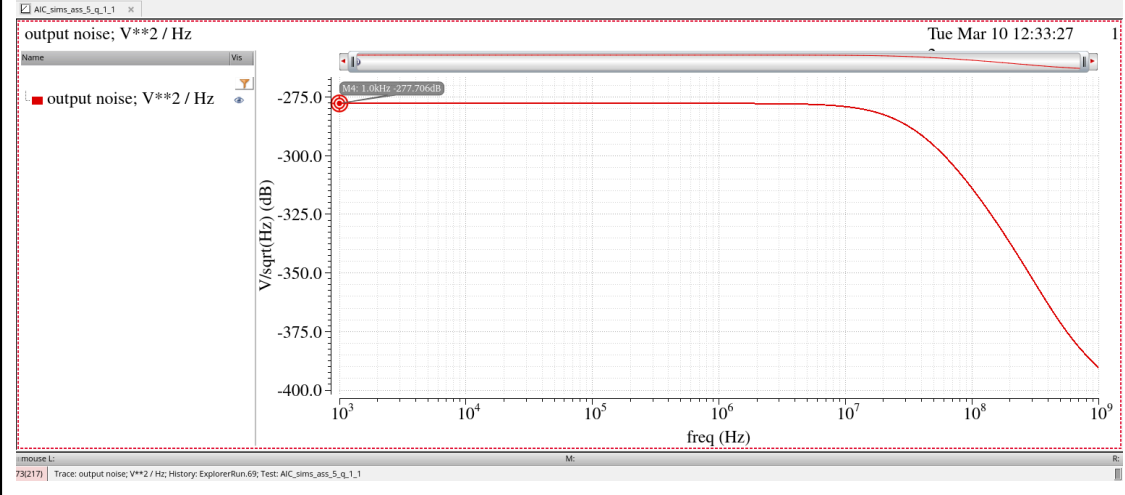


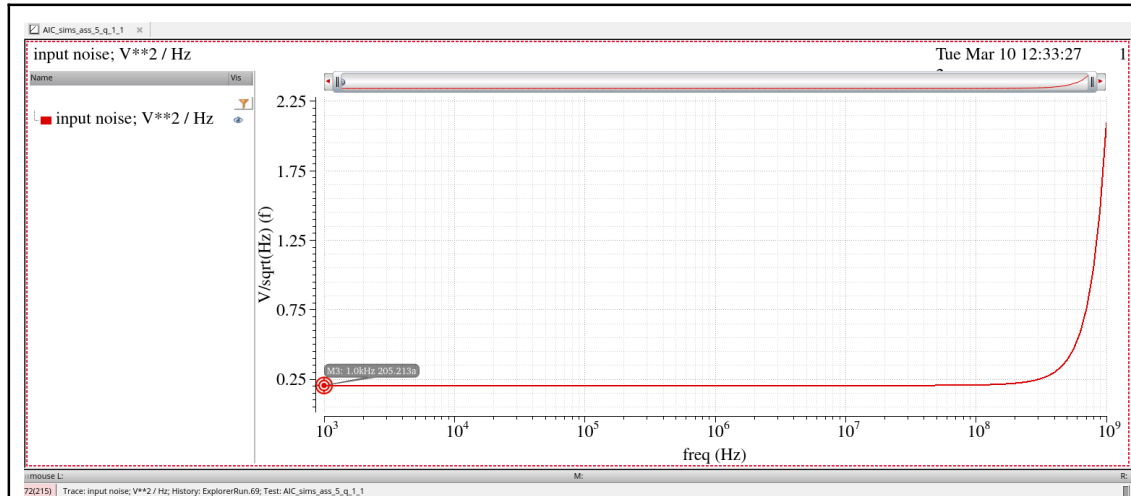


d)



e)





f)

Specification	Value
DC gain	18.025 dB = 7.966
3-dB bandwidth, f_B	22.98 MHz
Unity loop gain frequency	17.07 MHz
Phase margin	79.42°
Input Noise PSD (at low frequencies)	$205.21 \times 10^{-18} \text{ V}^2/\text{Hz}$

Component	% of noise contribution (low frequency)
R_i	80.77
R_f	10.10
G_{m1}	09.04
G_{m2}	00.00
R_L	00.00

The screenshot shows two overlapping windows from the Cadence software. The top window is the 'Noise Summary' dialog box, which is used to configure noise analysis parameters. It includes a title bar with a close button, a 'Print the output noise of `noise` analysis' button, and several configuration options: 'Type' (radio buttons for 'spot noise' and 'integrated noise', with 'integrated noise' selected), 'noise unit' (a dropdown menu showing 'V^2'), 'From (Hz)' (a text box with '1K'), 'To (Hz)' (a text box with '1M'), and 'weighting' (radio buttons for 'flat' and 'from weight file', with 'flat' selected). The bottom window is the 'Results Display Window', which has a title bar with a close button and a menu bar with 'Window', 'Expressions', 'Info', and 'Help'. The main content area of this window displays a table of noise contributors and a summary of the integrated noise results.

Noise Summary

Print the output noise of `noise` analysis

Type: ☐ spot noise ☒ integrated noise noise unit: V²

From (Hz): 1K To (Hz): 1M

weighting: ☒ flat ☐ from weight file

Results Display Window

Window Expressions Info Help

Device	Param	Noise Contribution	% Of Total
/RI	rn	1.05026e-08	80.77
/Rf	rn	1.3129e-09	10.10
/RN1	rn	1.17531e-09	9.04
/R01	rn	1.17531e-11	0.09
/RN2	rn	6.4769e-15	0.00
/Rload	rn	2.38976e-16	0.00
/Ro2	rn	6.4769e-17	0.00
/R9	rn	0	0.00
/R10	rn	0	0.00
/R11	rn	0	0.00
/R13	rn	0	0.00
/R14	rn	0	0.00
/R15	rn	0	0.00
/R7	rn	0	0.00

Integrated Noise Summary (in V²) Sorted By Noise Contributors
 Total Summarized Noise = 1.30026e-08
 Total Input Referred Noise = 2.05008e-10
 The above noise summary info is for noise data

Maestro window with simulation setup -

Setup

Name: **ass_5_q_1**

Analyses:

- ☒ tran 0 to conservative
- ☒ ac 1k 1G 20 Logarithmic Points Per D...
- ☒ noise 1k 1G 20 Logarithmic Points Per ...

Click to add analysis

Design Variables

- ☒ vt 20G
- ☒ A 8
- ☒ RB 20M
- ☒ CL 24p
- ☒ RL 1/(3.1416*HB*CL)
- ☒ Ri 10k
- ☒ Gm1 (k-1)*C<+2*3.1416*HB
- ☒ Cc 1p
- ☒ Gm2 0.04087

Click to add variable

☒ Parameters

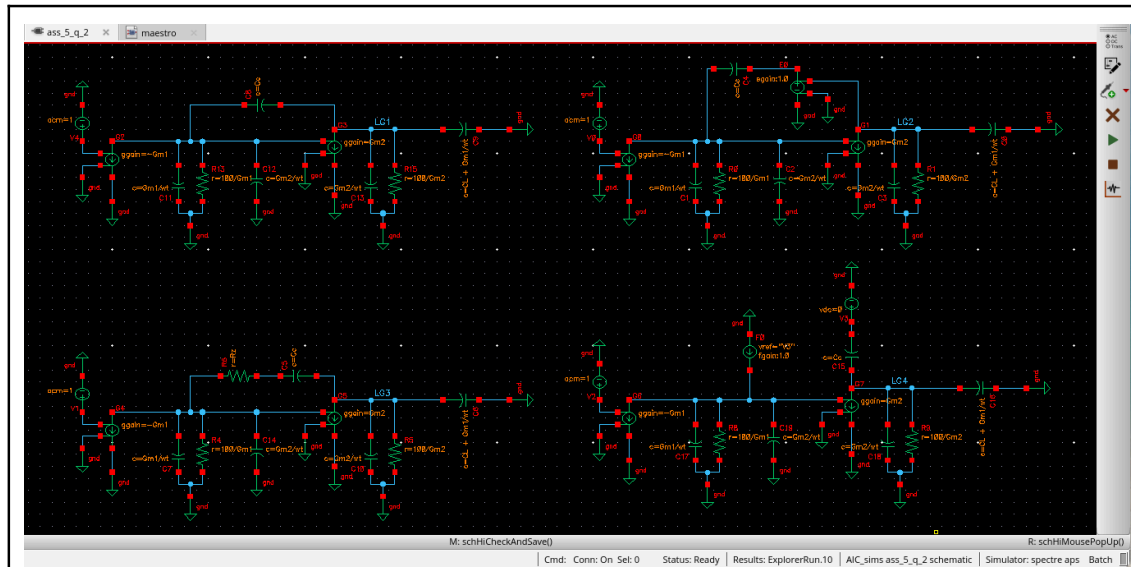
Name	Type	Details	Value	Plot	Save	Spec
v/LG_point; ac deg(V)	expr	phaseDegUnwrappedVfreq/ac "v/LG_point"]		☒	☐	
v/LG_point; ac dB20(V)	expr	dbVfreq/ac "v/LG_point"]		☒	☐	
v/Vout; ac dB20(V)	expr	dbVfreq/ac "vout"]		☒	☐	
v/Vin; tran (V)	expr	vtime(tran "vin")		☐	☐	
v/Vout; tran (V)	expr	vtime(tran "vout")		☐	☐	
output noise; V**2 / Hz	expr	db[getData("out")?result "noise"]**2]		☒	☐	
input noise; V**2 / Hz	expr	(getData("in")?result "noise")**2]		☒	☐	

Explorer Run Summary

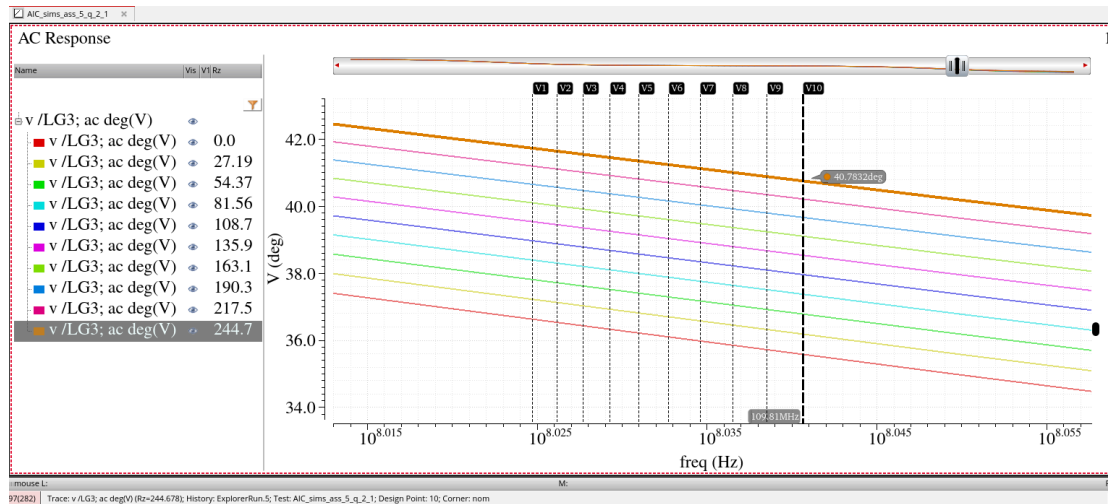
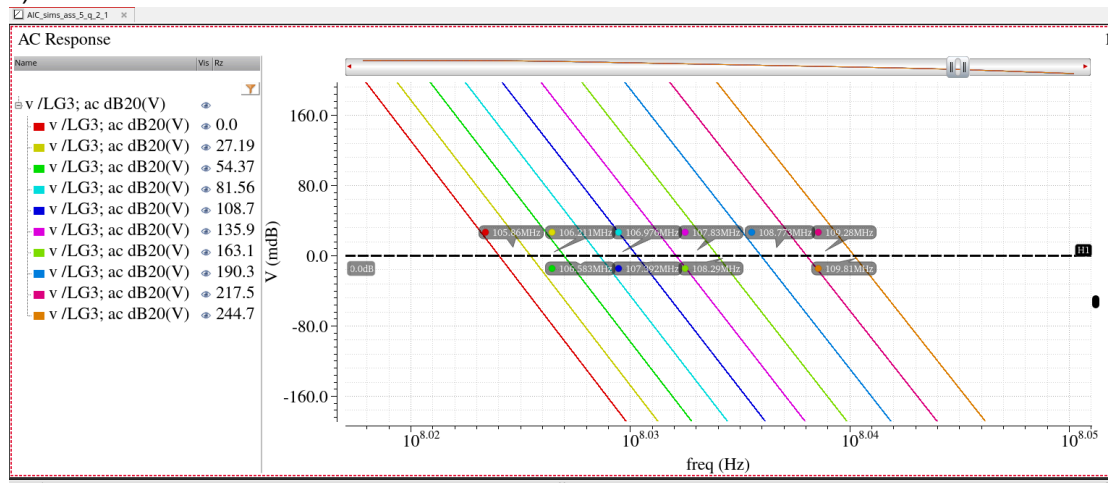
☒ 0 Corner

☒ Nominal Corner

5.2:



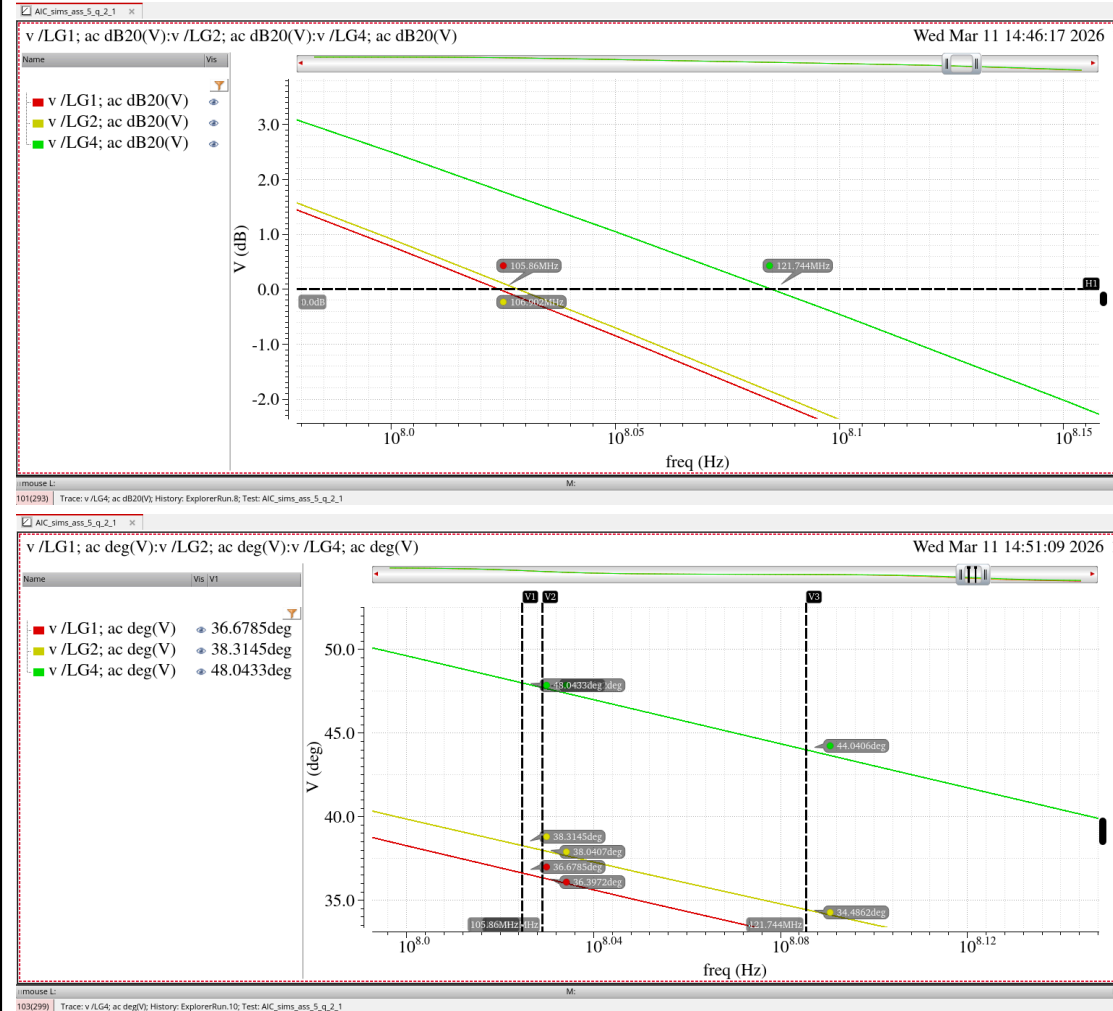
b)



Thus, we can conclude that the higher the Rz, the higher the phase margin. This is

because as R_z increases, the zero moves into the left half plane and keeps increasing the phase margin.

a, c, d)



The phase margins in c) and d) are better because the zero is removed and does not contribute to the phase margin anymore. The cross terms of C_c with C_1 and G_{01} in c) and C_2 and G_{02} in d) also disappear.

Case	Phase margin
a)	36.68°
b)	40.78°
c)	38.04°
d)	44.04°

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A2. a)

$$-G_{m1} v_T + v_1 (G_{01} + sC_1) + (v_1 - v_0) sC_c = 0$$

$$(v_1 - v_0) sC_c = G_{m2} v_1 + v_0 (G_{02} + sC_2)$$

$$\therefore v_1 = \frac{v_0 (G_{02} + sC_c + sC_2)}{sC_c - G_{m2}}$$

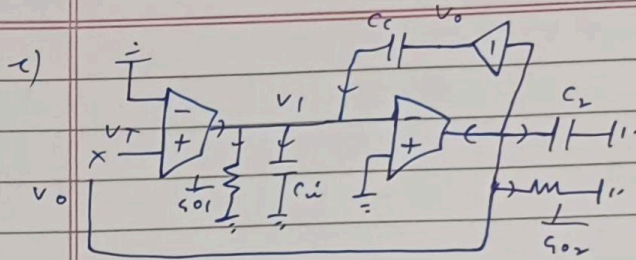
$$\therefore L(s) = A(s) = \frac{v_0}{v_T} = \frac{-G_{m1} (G_{m2} - sC_c)}{s^2 (C_1 C_c + C_2 C_c + C_1 C_2) + s [G_{01} C_2 + G_{02} C_1 + C_c (G_{m2} + G_{01} + G_{02})] + G_{01} G_{02}}$$

b) substituting $sC_c \rightarrow \frac{sC_c}{1 + sC_c R_2}$

$$L(s) = A(s) = \frac{v_0}{v_T} = \frac{-G_{m1} [G_{m2} + sC_c (G_{m2} R_2 - 1)]}{s^3 C_1 C_2 R_2 C_c + s^2 [C_c C_1 + C_c C_2 + C_1 C_2 + C_c R_2 (G_{02} C_1 + G_{01} C_2)] + s [C_c (G_{01} + G_{02} + G_{m2}) + G_{02} C_1 + G_{01} C_2 + C_c R_2 G_{01} G_{02}] + G_{01} G_{02}}$$

$$Z_{in} = \frac{1}{sC_c} \quad Z_{out} = \frac{1 + sC_c R_2}{sC_c}$$

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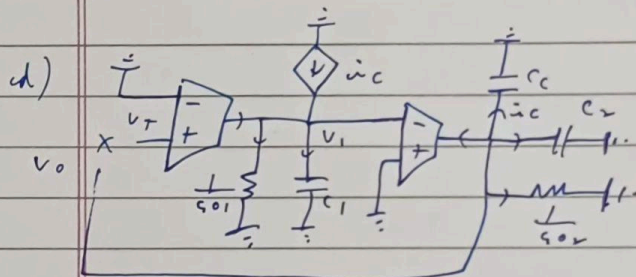


$$g_{m1} V_T + (V_0 - V_1) s C_c = V_1 (g_{01} + s C_1)$$

$$g_{m2} V_1 + V_0 (g_{02} + s C_2) = 0$$

$$\therefore V_1 = \frac{-V_0 (g_{02} + s C_2)}{g_{m2}}$$

$$\therefore \frac{V_0(s)}{V_T} = \frac{-g_{m1} g_{m2}}{s^2 (C_1 C_2 + C_c C_2) + s [g_{01} C_2 + g_{02} C_1 + C_c (g_{m2} + g_{02})] + g_{01} g_{02}}$$



$$\dot{v}_c = V_0 s C_c$$

$$g_{m1} V_T + V_0 s C_c = V_1 (g_{01} + s C_1)$$

$$g_{m2} V_1 + V_0 s C_c + V_0 (g_{02} + s C_2) = 0$$

$$\therefore A_4(s) = \frac{g_{m1} g_{m2}}{g_{01} g_{02} + s [g_{02} C_1 + g_{01} C_2 + C_c (g_{m2} + g_{01})] + s^2 (C_1 C_2 + C_c C_2)}$$

-
- c) Decouple C_c with C_1 and g_{01} and removes the pole.
 - d) Decouple C_c from C_2 and g_{02} and removes the pole.